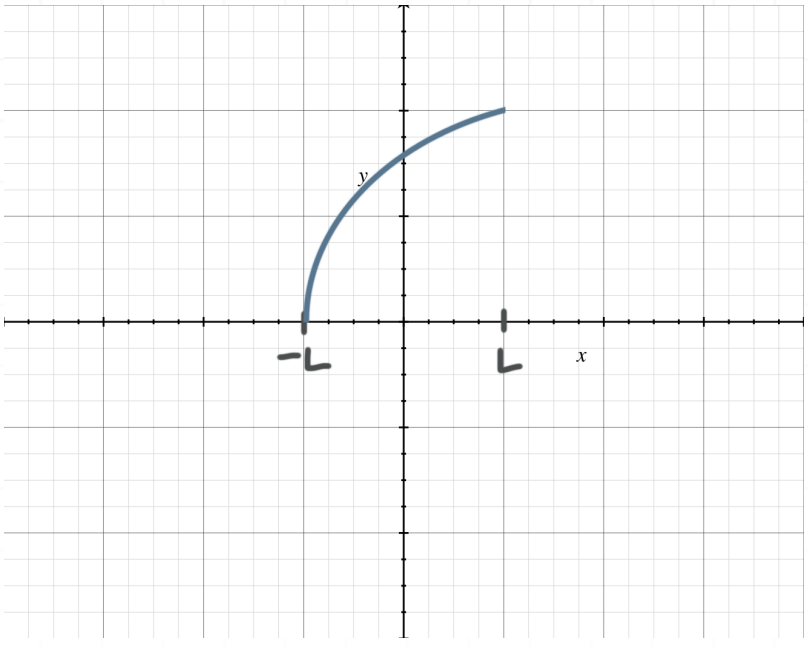
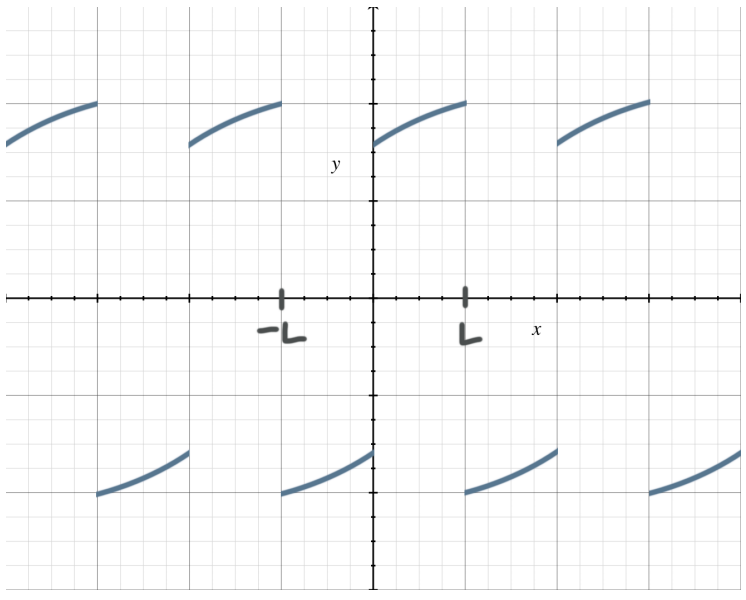
Topic: Periodic functions and periodic extensions

Question: Sketch the function's periodic extension.

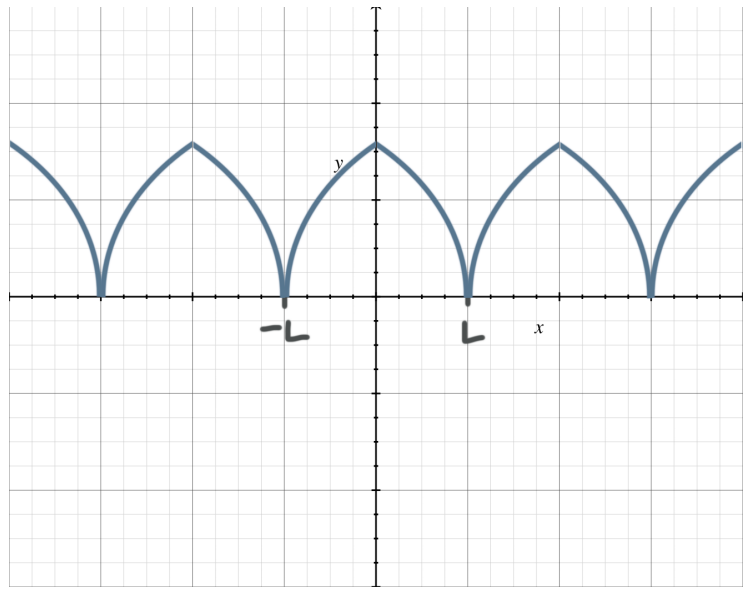


Answer choices:

A

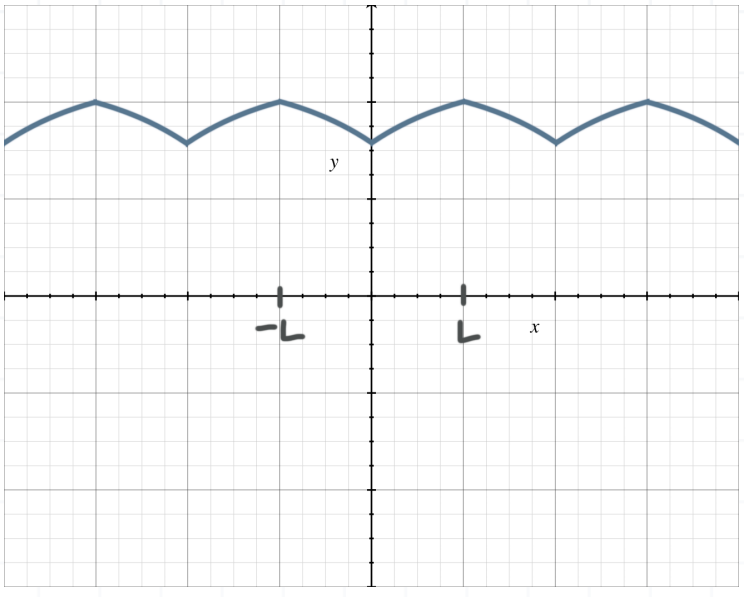


B

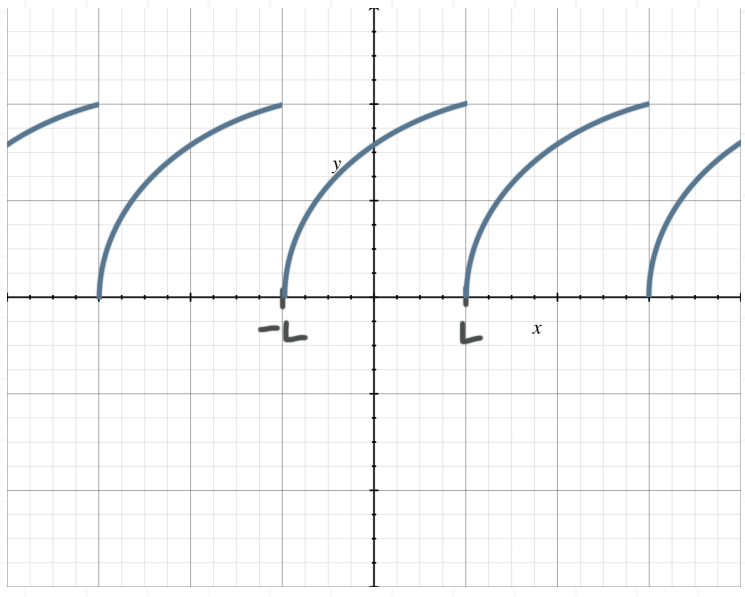


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C



D



Solution: D

The periodic extension of a function is the function we get when we repeat the portion of the graph on $-L \leq x \leq L$ over and over to both the left and right, on $(-\infty, -L)$ and on (L, ∞). Answer choice D is the graph of the periodic extension.

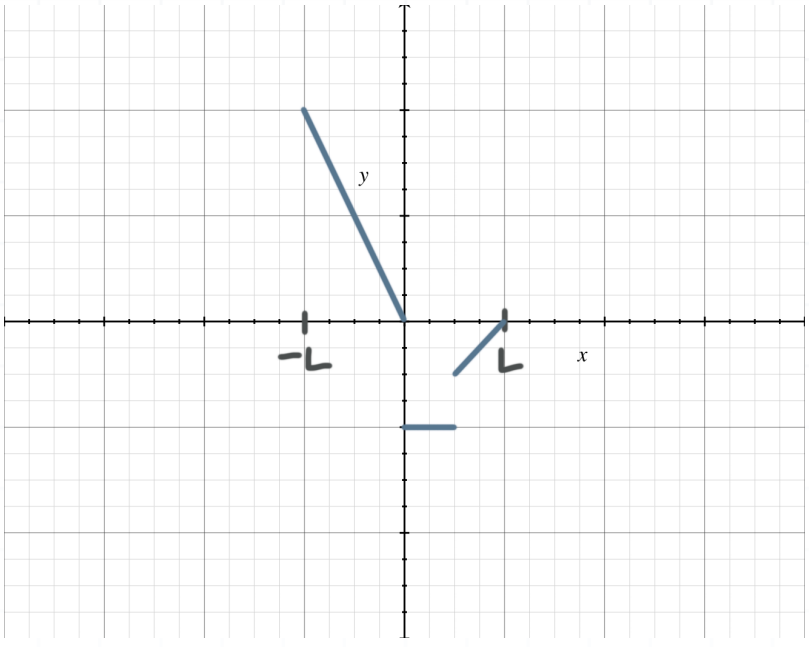
Answer choice A is the odd extension of the portion of the graph on $0 \leq x \leq L$, and answer choice C is the even extension of the portion of the graph on $0 \leq x \leq L$.

We could essentially say that answer choice B is the even extension of the portion of the graph on $-L \leq x \leq 0$, but that's not something we typically do.

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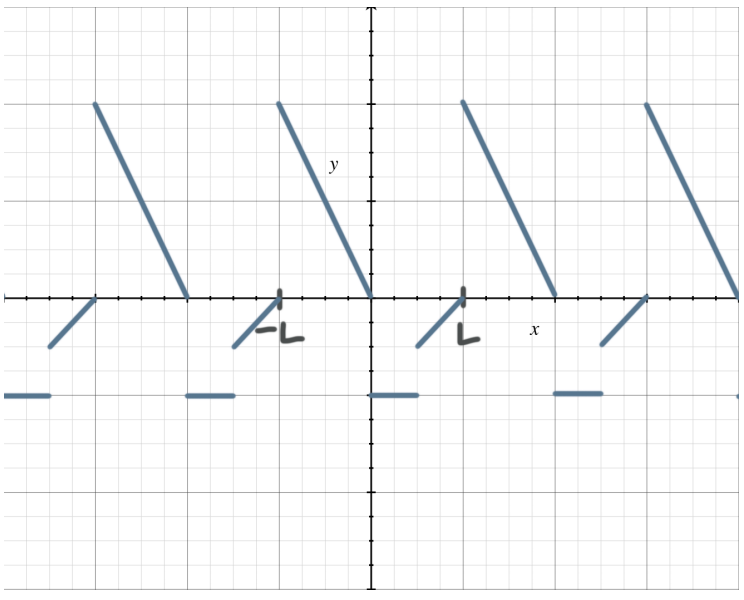
Topic: Periodic functions and periodic extensions

Question: Sketch the function's even extension.

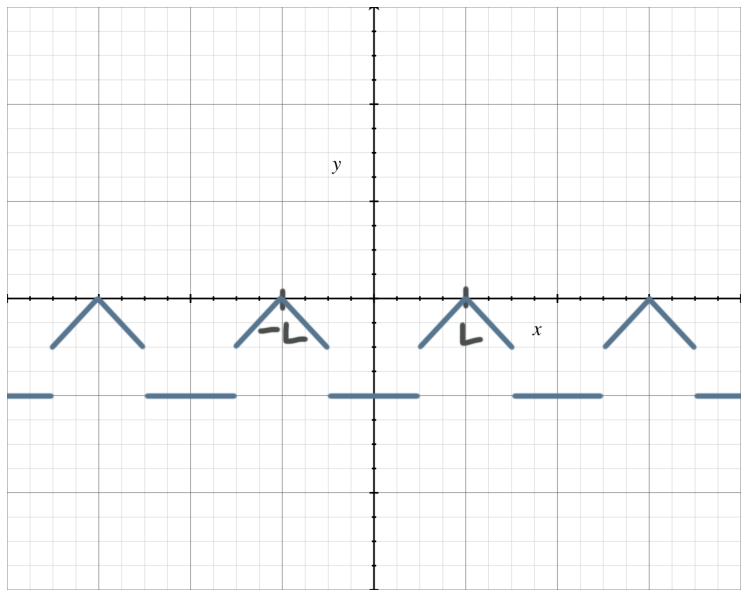


Answer choices:

A

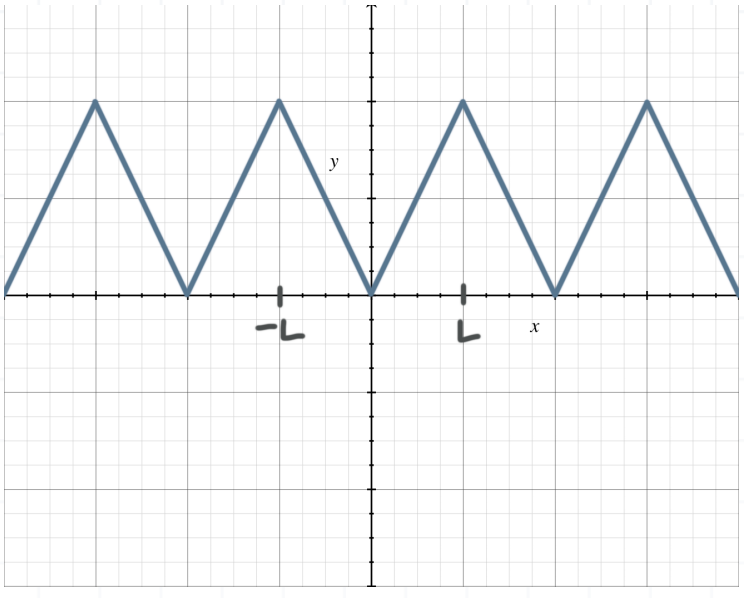


B

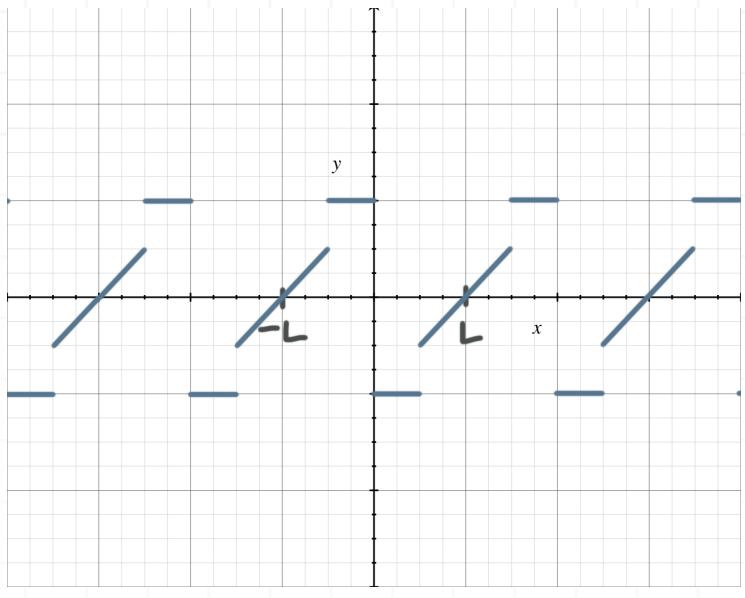


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C



D



Solution: B

The even extension of a function is the function we get when we reflect the portion of the graph on $0 \leq x \leq L$ into the interval $-L \leq x \leq 0$, and then repeat that symmetrical graph over and over to both the left and right, on $(-\infty, -L)$ and on (L, ∞). Answer choice B is the graph of the even extension.

Answer choice A is the periodic extension of the portion of the graph on $-L \leq x \leq L$, and answer choice D is the odd extension of the portion of the graph on $0 \leq x \leq L$.

We could essentially say that answer choice C is the even extension of the portion of the graph on $-L \leq x \leq 0$, but that's not something we typically do.

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Topic: Periodic functions and periodic extensions

Question: Find the even and odd extensions of the function.

$$f(x) = 1 + x + x^2$$

Answer choices:

A $$E(x) = \begin{cases} 1 - x + x^2 & 0 \le x \le L \\ 1 + x + x^2 & -L \le x < 0 \end{cases} \text{ and } O(x) = \begin{cases} -1 + x - x^2 & 0 \le x \le L \\ 1 + x + x^2 & -L \le x < 0 \end{cases}$$

B $$E(x) = \begin{cases} 1 + x + x^2 & 0 \le x \le L \\ 1 - x + x^2 & -L \le x < 0 \end{cases} \text{ and } O(x) = \begin{cases} 1 + x + x^2 & 0 \le x \le L \\ -1 + x - x^2 & -L \le x < 0 \end{cases}$$

C $$E(x) = \begin{cases} -1 + x - x^2 & 0 \le x \le L \\ 1 + x + x^2 & -L \le x < 0 \end{cases} \text{ and } O(x) = \begin{cases} 1 - x + x^2 & 0 \le x \le L \\ 1 + x + x^2 & -L \le x < 0 \end{cases}$$

D $$E(x) = \begin{cases} 1 + x + x^2 & 0 \le x \le L \\ -1 + x - x^2 & -L \le x < 0 \end{cases} \text{ and } O(x) = \begin{cases} 1 + x + x^2 & 0 \le x \le L \\ 1 - x + x^2 & -L \le x < 0 \end{cases}$$

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Solution: B

The even extension of $f(x) = 1 + x + x^2$ is

$$E(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L \leq x < 0 \end{cases}$$

$$E(x) = \begin{cases} 1 + x + x^2 & 0 \leq x \leq L \\ 1 - x + (-x)^2 & -L \leq x < 0 \end{cases}$$

$$E(x) = \begin{cases} 1 + x + x^2 & 0 \leq x \leq L \\ 1 - x + x^2 & -L \leq x < 0 \end{cases}$$

The odd extension of $f(x) = 1 + x + x^2$ is

$$O(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$O(x) = \begin{cases} 1 + x + x^2 & 0 \leq x \leq L \\ -(1 - x + (-x)^2) & -L \leq x < 0 \end{cases}$$

$$O(x) = \begin{cases} 1 + x + x^2 & 0 \leq x \leq L \\ -1 + x - x^2 & -L \leq x < 0 \end{cases}$$

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